HW#5. MATH 464. Lazizbek.

Determine whether or not the following are polyhedra. Use clear reasoning & appeal to the definitions & theorems from Chapter 2 of the textbook.

Pr 1.

$$\Omega_1 = \emptyset.$$

Solution:

Let

(1) $\Omega_1 = \emptyset.$

I claim that this Ω_1 is a polyhedron, because it satisfies the definition of a polyhedron which I present below along with the proof.

Def. 2.1.

A **polyhedron** is a set of the form

(2) $P = \{ x \in \mathbb{R}^n \mid Ax \geq b \}$

where A is an $m \times n$ matrix, and $b \in \mathbb{R}^m$.

Remark 1 (to Def. 2.1.)

Any polyhedron can be written as a <u>finite</u> intersection of halfspaces.

Using this definition & remark, let me now try to prove Ω_1 is a polyhedron:

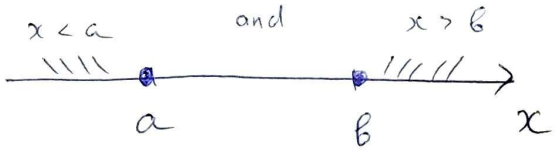
Proof.

Let me write down this set $\Omega_1 = \emptyset$ in a modified way:

(3) $\Omega_1 = \{x \in \mathbb{R} \mid x \leq a, \ x \geq b, \ a < b\}$

Pictorially

(4)



If we pay attention to this picture (4), we see that $\nexists$ x s.t $x \leq a$, $x \geq b$ for $a < b$, i.e.

(5) $\{x \leq a\} \cap \{x \geq b\} = \emptyset$ for $a < b$.

So

(6) $\Omega_1 = \emptyset = \{x \in \mathbb{R} \mid x \leq a, \ x \geq b, \ a < b\}$.

Now, let me try this set notation similar to the one in the definition by creating matrix A & vector f.

So for $a < b$

(7) $\begin{cases} x \leq a \\ x \geq b \end{cases} \cdot (-1) \Rightarrow \begin{cases} -x \geq -a \\ x \geq b \end{cases} \Rightarrow$

(8) $\begin{bmatrix} -1 \\ 1 \end{bmatrix} x \geq \begin{bmatrix} -a \\ b \end{bmatrix}$ for $a < b$.

So, we can write (6) by this matrix notation (8) as follows:

(9) $\emptyset = \{x \in \mathbb{R} \mid x \leq a,\ x \geq b,\ a < b\} =$

$$= \{x \in \mathbb{R} \mid \underbrace{\begin{bmatrix} -1 \\ 1 \end{bmatrix}}_{A} x \geq \underbrace{\begin{bmatrix} -a \\ b \end{bmatrix}}_{f},\ a < b\} =$$

$$= \{x \in \mathbb{R} \mid Ax \geq f,\ a < b\}$$

where $A = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ & $f = \begin{bmatrix} -a \\ b \end{bmatrix}$, which is corresponding to the def. 2.1. above, hence $\Omega_1 = \emptyset$ is a polyhedron.

$\blacksquare$

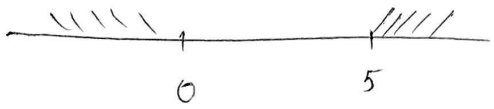
Pr 2.

(10) $\Omega_2 = \{x \in \mathbb{R} \mid x \geq 5,\ x \leq 0\}$

Solution.

Actually, I've found a way for Pr.1 after this Pr.2.

$\Omega_2 = \emptyset$ if we make a picture of set (10):



(11) $\{x \geq 5\} \cap \{x \leq 0\} = \emptyset$.

So, $\Omega_2 = \emptyset$, hence by the result from Pr. 1, Ω_2 is a polyhedron.

Also another method would be forcing set Ω_2 to match def 2.1. on Page 1:

$$\Omega_2 = \left\{x \in \mathbb{R} \;\middle|\; \begin{matrix} x \geq 5 \\ x \leq 0 \end{matrix} \right\} \Rightarrow \left\{x \in \mathbb{R} \;\middle|\; \begin{matrix} x \geq 5 \\ -x \geq 0 \end{matrix} \right\} =$$

$$= \left\{x \in \mathbb{R} \;\middle|\; \begin{bmatrix} 1 \\ -1 \end{bmatrix} x \geq \begin{bmatrix} 5 \\ 0 \end{bmatrix} \right\} = \{x \in \mathbb{R} \mid Ax \geq b\}$$

where $A = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ & $b = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$, hence Ω_2 is a polyhedron.

$\blacksquare$

Pr. 3.

(12) $\Omega_3 = \{x \in \mathbb{R} \mid x^2 - 9 \leq 0\}$

Solution:

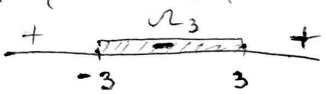
Let

(12) $\Omega_3 = \{x \in \mathbb{R} \mid x^2 - 9 \leq 0\}$

and I claim this set Ω_3 is a polyhedron, since it satisfies the definition of a polyhedron, def. 2.1.

WTS: $\Omega_3 = \{x \in \mathbb{R} \mid x^2 - 9 \leq 0\} = \{x \in \mathbb{R} \mid Ax \geq b\}$.

(13) .....



Again, using similar technique in my previous solutions, let me try to force Ω_3 look like the RHS:

$$\Omega_3 = \{x \in \mathbb{R} \mid x^2 - 9 \leq 0\} = \{x \in \mathbb{R} \mid x \geq -3,\ x \leq 3\} =$$

$$= \{x \in \mathbb{R} \mid \left.\begin{cases} x \geq -3 \\ x \leq 3 \end{cases}\right\} \cdot (-1) = \left\{x \in \mathbb{R} \;\middle|\; \begin{matrix} x \geq -3 \\ -x \geq -3 \end{matrix}\right\} =$$

$$= \left\{x \in \mathbb{R} \;\middle|\; \underbrace{\begin{bmatrix} 1 \\ -1 \end{bmatrix}}_{A} x \geq \underbrace{\begin{bmatrix} -3 \\ -3 \end{bmatrix}}_{b}\right\} = \{x \in \mathbb{R} \mid Ax \geq b\}$$

where $A = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $b = \begin{bmatrix} -3 \\ -3 \end{bmatrix}$, so

Ω_3 is a polyhedron using def. 2.1.

Pr4.

(14) $\Omega_4 = \{ x \in \mathbb{R}^n \mid \|x\| \leq 1 \}$.

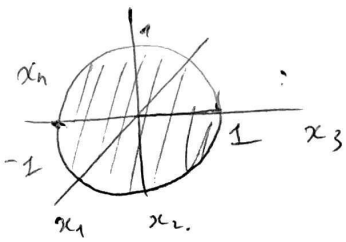
Solution:

Let

(14) $\Omega_4 = \{ x \in \mathbb{R}^4 \mid \|x\| \leq 1 \}$. $\Rightarrow$

(15) $x_1^2 + x_2^2 + \dots + x_n^2 \leq 1$, and pictorially

(16)

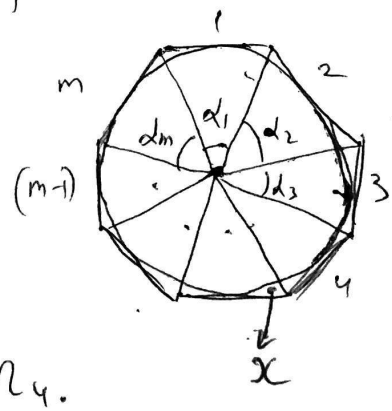


I claim this disc, Ω_4, is not a polyhedron as it can never be written as an intersection of finite number of halfspaces in attempt to use the Remark 1 for def. 2.1 op Page 1.

For contradiction, let $\Omega_4 = P = \{ x \in \mathbb{R}^n \mid Ax \geq b \}$

Let m be the maximum number of halfspaces indexed as $1, 2 \dots m$ making the entries of A & b in def. 2.1:

(17) $A = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix}$ $b = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$



So $\{ Ax \geq b \mid x \in \mathbb{R}^n \} = P = \Omega_4$.

But we can see in image (17), it does not matter how many (large amount of) half spaces (1, 2 ... m) we consider in P, there's always some points like, red x s.t. $x \in P$, $x \notin \Omega_4$, which are not entirely mapped to the disc. The reason for this is connected to central angles α_i. For simplicity, let

(18) $$\alpha_1 = \alpha_2 = \dots \alpha_m = \alpha.$$

For this disc to fully fit inside K ($K \leq m$) half spaces, very closely $\alpha \approx 0°$, i.e.

(19) $$\alpha = \frac{360°}{K} \to 0°. \text{ must be satisfied;}$$

in limit language:

(20) $$\lim_{K \to m} \alpha = \lim_{K \to m} \frac{360°}{K} = \frac{360°}{m} \neq 0 \quad \text{for}$$

finite m, if we consider maximum number of halfspaces to be at most m, which is contradiction!

For (19) to be satisfied, $K \to \infty$ is clear.

$\Rightarrow$

(21) $\lim_{K \to \infty} \frac{360^\circ}{K} = \frac{360^\circ}{\infty} = 0^\circ.$

So, $m = \infty$, we need infinite number of halfspaces, hence by Remark 1, on Page 1, Ω_4 is not a polyhedron. $\blacksquare$